

Assignment 4b: Risk-Based Allocation on the Euro Stoxx 50

Group 12

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Part I: Equal Risk Contribution (ERC)

The fundamental advantage of the Equal Risk Contribution portfolio is that it guarantees that each asset contributes exactly the same amount of marginal risk to the total volatility. Let us implement it in this First Part.

Mathematical Formulation and Euler's Theorem

Let $\mathbf{w}$ be the vector of portfolio weights and Σ the covariance matrix of asset returns. The total portfolio volatility is $\sigma(\mathbf{w}) = \sqrt{\mathbf{w}^T \Sigma \mathbf{w}}$. Because volatility is a homogeneous function of degree 1, we can apply Euler's theorem to decompose the total risk into a linear combination of individual Marginal Risk Contributions (MRC):

$$\sigma(\mathbf{w}) = \sum_{i=1}^N w_i \frac{\partial \sigma(\mathbf{w})}{\partial w_i} = \sum_{i=1}^N \frac{w_i (\Sigma \mathbf{w})_i}{\sqrt{\mathbf{w}^T \Sigma \mathbf{w}}} \quad (1)$$

The Total Risk Contribution (TRC) of asset i is defined as $\text{TRC}_i = w_i \frac{(\Sigma \mathbf{w})_i}{\sigma(\mathbf{w})}$. The ERC portfolio requires that $\text{TRC}_i = \text{TRC}_j$ for all pairs (i, j) , meaning each asset contributes exactly $\sigma(\mathbf{w})/N$ to the overall portfolio risk.

Numerical Implementation

Because a closed-form solution for ERC does not exist when correlations are non-zero, the weights must be found by solving a non-linear optimization problem. We formulated the objective function to minimize the variance of the risk contributions:

$$\mathbf{w}_{\text{ERC}} = \arg \min_{\mathbf{w}} \sum_{i=1}^N \sum_{j=1}^N (w_i (\Sigma \mathbf{w})_i - w_j (\Sigma \mathbf{w})_j)^2 \quad \text{s.t.} \quad \sum_{i=1}^N w_i = 1, w_i \geq 0 \quad (2)$$

Note that without proper scaling, the gradients computed by the SLSQP optimizer fall below machine tolerance, causing premature convergence. To solve this, we normalized the covariance matrix by dividing it by the mean of its diagonal elements:

$$\Sigma_{\text{norm}} = \frac{\Sigma}{\frac{1}{N} \sum_{i=1}^N \Sigma_{i,i}} \quad (3)$$

This normalization significantly improves the solver's precision.

Backtest Results

We backtest the ERC portfolio for stocks in the Euro Stoxx 50, from January 2015 to February 2023. We compare the results for different covariance estimation methods: sample, constant correlation and

market factor (cf. Assignment 3). As a benchmark we implemented an equally weighted (buy-and-hold) portfolio.

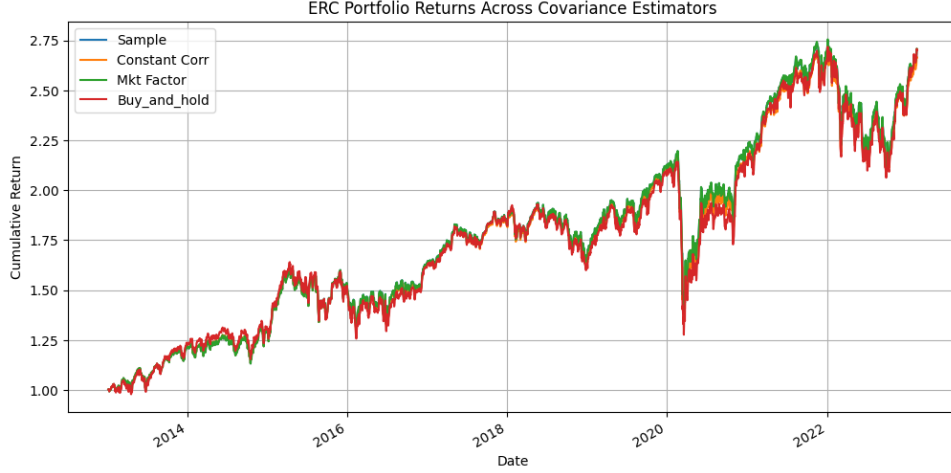


Figure 1: Cumulative performance of different strategies, 10 bps transaction costs.

Table 1: ERC Portfolio Metrics (10 bps transaction costs)

Method	Cumul. Return	Ann. Vol (EWM63)	Max Drawdown	Avg Turnover
Sample	2.7028	16.34%	−37.27%	0.0243
Constant Corr	2.6640	16.45%	−38.20%	0.0227
Mkt Factor	2.7580	16.33%	−37.21%	0.0241
Buy & Hold	2.7014	17.56%	−40.40%	0.0008

The three ERC variants produce very similar cumulative returns (≈ 2.66 – 2.76). The constant-correlation estimator delivers the lowest turnover (0.0227) thanks to its stabilising structure. All ERC portfolios reduce volatility (~ 16.3 – 16.4%) and drawdown (~ 37.2 – 38.2%) relative to the Buy & Hold benchmark (17.56% vol, -40.40% DD). The buy-and-hold portfolio achieves a comparable cumulative return (2.70) but at the cost of slightly higher risk. Also note that the non-zero turnover of the equally weighted portfolio is due to some companies being replaced in the Stoxx 50.

Part II: Hierarchical Risk Parity (HRP)

The Three-Stage Algorithm

The HRP algorithm consists of three sequential steps:

1. **Tree Clustering:** First, we map the correlation matrix into a distance metric space using $d_{i,j} = \sqrt{0.5(1 - \rho_{i,j})}$. This ensures that highly correlated assets have distances close to zero. We then apply an agglomerative hierarchical clustering algorithm (e.g., Ward linkage) to group assets into a dendrogram.
2. **Quasi-Diagonalization:** The covariance matrix is reorganized topologically based on the order of the leaves in the dendrogram. This step places similar assets adjacent to each other, highlighting the block-diagonal structure of the market.
3. **Recursive Bisection:** Starting from the root of the tree, capital is allocated top-down. At each node, the allocation is split between the left (L) and right (R) branches based on inverse variance parity. The split factor α is:

$$\alpha = 1 - \frac{\sigma_L^2}{\sigma_L^2 + \sigma_R^2} \quad (4)$$

where σ_L^2 and σ_R^2 are the estimated variances of the left and right sub-clusters.

PCA Detoning Intuition

Financial correlation matrices are typically dominated by the first principal component, which represents the overarching “market factor”. This factor tends to push all pairwise correlations upward, masking the true underlying sector relationships. By removing the loudest signal (the market), we hope that the hierarchical clustering algorithm will accurately identify the idiosyncratic clusters, leading to a much more granular and effective diversification tree.

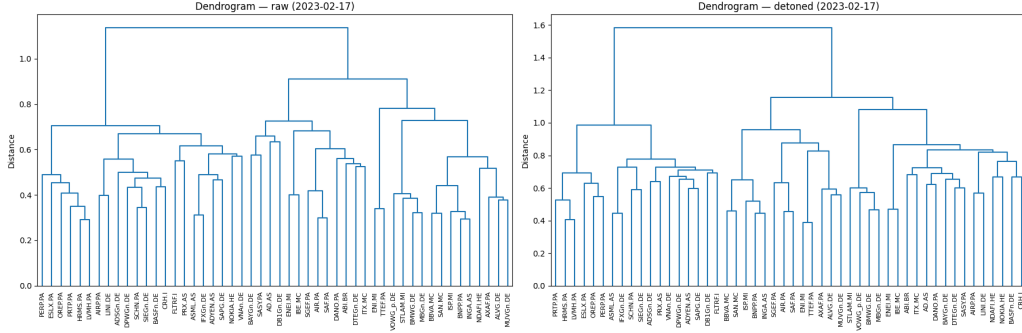


Figure 2: Correlation Matrix without and with detoning

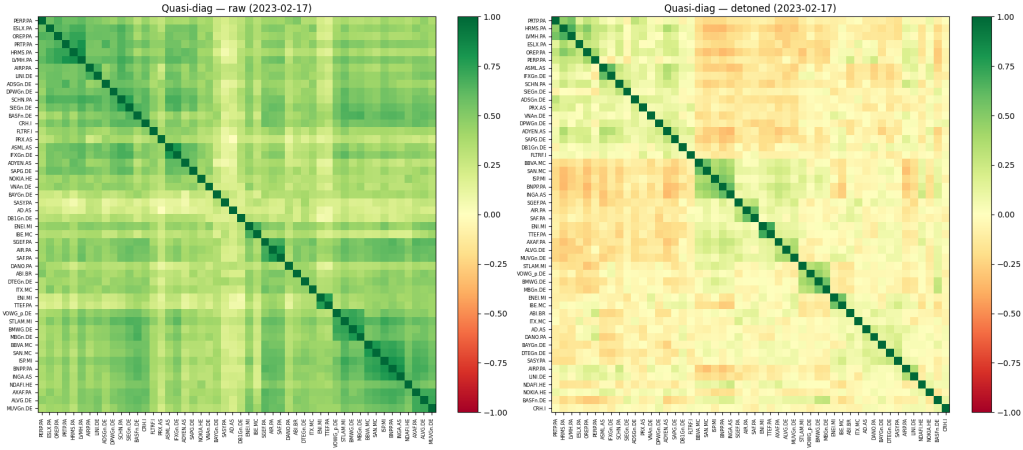


Figure 3: Correlation Matrix without and with detoning

Cluster Stability and the Rand Index

To formally evaluate the benefit of detoning, we measured cluster stability across consecutive rebalancing dates using the **Rand Index**. The Rand Index computes the similarity between two data clusterings by counting the fraction of pairs that are assigned to the same or different clusters consistently.



Figure 4: Rand Index

The data is smoothed using a 12-month Exponentially Weighted Moving Average (EWM). Structured covariance estimators, particularly when subjected to market detoning (dashed lines), demonstrate significantly higher temporal consistency compared to the standard empirical sample covariance (solid blue line), which remains highly susceptible to market noise. Furthermore, the overall temporal stability shows an upward trend as the number of clusters increases.

Sensitivity of HRP Weights

We run HRP across the full grid of pipeline choices $\{\text{linkage method}\} \times \{\text{allocation traversal}\} \times \{\text{covariance estimator}\} \times \{\text{detoning on/off}\}$ and summarise each variant by its mean turnover and mean Herfindahl–Hirschman Index (HHI), where:

$$\text{HHI}(\mathbf{w}) = \sum_i w_i^2 \quad (\min 1/N \approx 0.02 \text{ for } 50 \text{ assets; max } 1).$$

Table 2: Mean Turnover and Mean HHI across the HRP grid (averaged over estimators and detoning).

Linkage	Allocator	Mean TO	Mean HHI
average	dendrogram bisection	0.1408	0.1186
average	recursive iteration	0.1020	0.0270
complete	dendrogram bisection	0.1833	0.0642
complete	recursive iteration	0.1157	0.0266
single	dendrogram bisection	0.1388	0.1724
single	recursive iteration	0.1018	0.0270
ward	dendrogram bisection	0.1805	0.0426
ward	recursive iteration	0.1112	0.0266

How does the choice of linkage propagate to the weights?

Recursive iteration produces stable, diversified allocations across all linkage methods: the HHI stays close to the minimum (~ 0.027) regardless of whether we use single, ward, complete or average linkage. The turnover also remains in a remarkably tight range (0.1018–0.1157). This is because it mechanically halves the leaf order, smoothing out the asymmetries in the dendrogram.

Dendrogram bisection, by contrast, faithfully follows the merge hierarchy and therefore amplifies the structural differences between linkage methods. Single linkage produces the highest concentration

(HHI = 0.172) due to chaining the algorithm repeatedly peels off isolated assets, creating highly asymmetric splits that allocate disproportionate capital to low-variance outliers. Ward linkage keeps HHI moderate (0.043) since it minimises within-cluster variance at each merge, producing balanced clusters. Complete (0.064) and average (0.119) fall between these extremes.

How do the two allocation traversals differ?

The two traversals produce similar weights when the dendrogram is naturally balanced (as with Ward linkage). They diverge when the dendrogram is asymmetric: recursive iteration mechanically halves the leaf list regardless of the tree structure, which produces more diversified but less economically meaningful splits; dendrogram bisection follows the actual merge hierarchy, which with single linkage can allocate disproportionate capital to isolated low-variance outliers.

Recursive iteration consistently achieves lower turnover across all linkage methods. This happens because halving the leaf order is a deterministic operation that changes smoothly when the leaf ordering shifts slightly from one rebalance to the next. Dendrogram bisection is more sensitive to small changes in the correlation structure because the identity of the “largest merge distance” cluster can flip between dates.

Part III: Connecting the Dots

3.a Robustness to Covariance Estimators (L1 Distance)

We stress-tested both ERC and HRP against three covariance estimators: the standard Sample Covariance, the Ledoit-Wolf Constant-Correlation Shrinkage, and the Single-Factor (Market) Shrinkage. As a representative HRP variant we chose Ward linkage with dendrogram iteration and no detoning. To measure robustness, we computed the **Average L1 distance** between the weight vectors produced across different estimators:

$$L1_{\text{dist}} = \sum_{i=1}^N |w_{\text{estimator } 1, i} - w_{\text{estimator } 2, i}| \quad (5)$$

Table 3: Average L1 Distance Between Portfolio Pairs

Pair	ERC	HRP
(sample, constant_corr)	0.0331	0.0328
(sample, mkt_factor)	0.0056	0.2855
(constant_corr, mkt_factor)	0.0341	0.2976

Comparing these variations reveals a stark contrast in robustness. **Equal Risk Contribution (ERC)** is highly stable. Its weights barely change regardless of the estimator used, with a maximum L1 distance of just 0.034 and a minimum of 0.006. Because ERC directly targets an equal risk contribution constraint mathematically, it converges to a similar equilibrium regardless of minor matrix adjustments.

3.b Performance Comparison: ERC vs HRP

Table 4: Portfolio Performance Metrics (10 bps transaction costs)

Strategy	Ann. Vol EWM63 (%)	Sharpe	Max Drawdown (%)	Avg Turnover
ERC sample	16.26	0.623	−37.24	0.0250
HRP sample	15.15	0.624	−31.08	0.3700
ERC constant_corr	16.38	0.609	−38.19	0.0218
HRP constant_corr	15.26	0.615	−32.07	0.3722
ERC mkt_factor	16.25	0.623	−37.18	0.0250
HRP mkt_factor	14.83	0.715	−30.36	0.3701
Buy and Hold	17.48	0.587	−40.40	0.0017

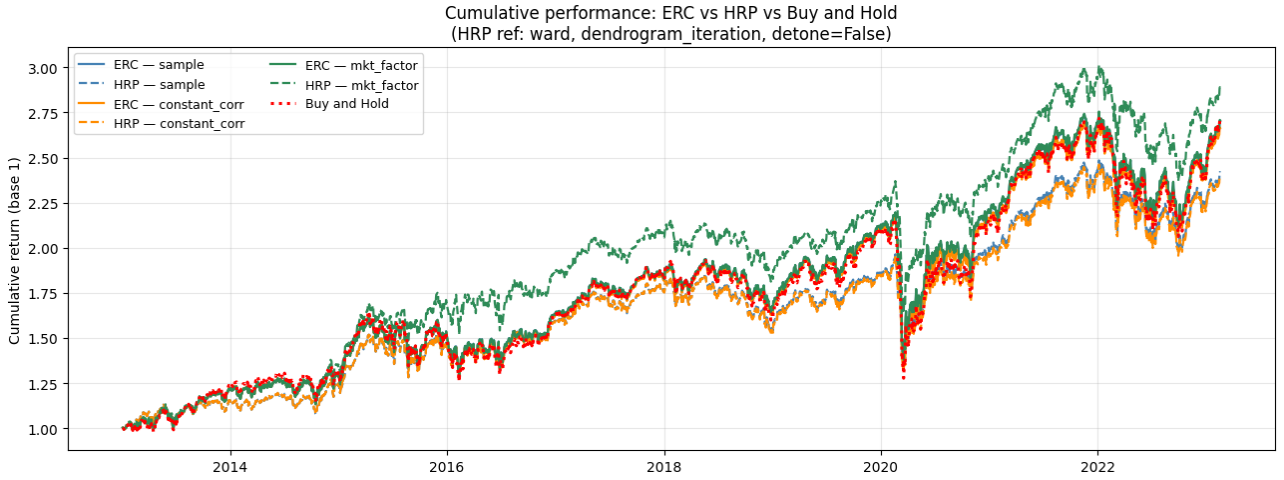


Figure 5: Cumulative returns comparison across strategies.

Comparing ERC and HRP reveals clear structural trade-offs:

- **Lower Risk (HRP):** HRP offers a more conservative risk profile. It achieves consistently lower realized volatility (ranging from 14.83% to 15.26% vs. ERC’s ~16.3%) and better mitigates market drops, limiting the Max Drawdown to −30.36% (*mkt_factor*) compared to ERC’s −37.24%. Both significantly outperform the Buy and Hold baseline in risk management.
- **Risk-Adjusted Returns:** Thanks to its superior drawdown control, HRP with the market factor achieves the highest Sharpe ratio overall (0.715), outperforming ERC (0.623). However, ERC remains more consistent in its Sharpe ratios across different estimators.
- **Higher Turnover (ERC wins):** HRP’s major drawback is high turnover. While ERC adjusts its weights smoothly over time (average turnover of roughly 0.022–0.025), HRP incurs a turnover roughly 15 times higher (around 0.37). Small correlation changes force HRP to significantly rebalance entire clusters.

3.c The Role of Shrinkage in Risk-Based vs Optimization-Based Allocations

Unlike classic *Mean-Variance Optimization* (MVO) approached in previous laboratories, shrinkage techniques (like *constant-correlation* or *mkt_factor*) barely affect and sometimes slightly degrade the performance of ERC and HRP.

This highlights a core algorithmic difference: Markowitz optimization requires inverting the covariance matrix. If the empirical data is noisy, inversion acts as an “error maximizer,” making shrinkage absolutely necessary to regularize the matrix and stabilize the weights.

Risk-based methods (ERC and HRP) do not invert the matrix in the same direct, unconstrained way. HRP uses it only to measure relative distances between assets, and ERC uses it iteratively to

balance marginal risks. Because these methods are naturally diversified and algorithmically robust to estimation errors, adding shrinkage provides only marginal benefits compared to pure mathematical optimization.

Conclusion

The difference between Lab 3 and Lab 4b lies in identifying who these strategies are designed for. Lab 3 focuses on pure returns while largely ignoring volatility, whereas in Lab 4b the goal is to reduce annualized volatility and drawdowns as much as possible.

Lab 3 targets hedge funds or aggressive asset managers, while Lab 4b targets pension funds. The fundamental difference lies in how performance is measured and compensation structured. A hedge fund like Citadel, which posted a 38% net return in 2023, is judged on absolute performance and compensates its traders primarily through P&L. A pension fund like Fidelity's retirement division, managing over \$2 trillion in assets, must instead track the market steadily while preserving capital, as volatility is an existential risk, not an opportunity.